Possible issue in pegasos classifer: Not training for data dimension besid

https://github.com/qiskit-community/qiskit-machine-learning/issues/344

ROW 82

Possible issue in pegasos classifer: Not training for data dimension besides 2 #344

New issue

Closed #345

charmerDark opened on Mar 10, 2022

Environment

- **Qiskit Machine Learning version:** 0.3.1
- **Python version:** 3.7.12
- **Operating system:** Ubuntu 18.04

What is happening?

The `PegasosQSVC` method seems to have issues taking in data of any format except (num_samples, 2). The fit method throws errors when using any other feature size. However the documentation mentions the arguments for `pegaus_qsvc.fit` being `(num_samples, num_fetaures)` and does not limit it to 2.

How can we reproduce the issue?

Taken from the [pegaus tutorial]

```
from sklearn.datasets import make_blobs
features, labels = make_blobs(n_samples=20, n_features=4, centers=2, random_state=3, shuffle=True)

import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler

train_features, test_features, train_labels, test_labels = train_test_split(features, labels, train_size=15, s

num_qubits = 4
tau = 100
C = 1000

from qiskit import BasicAer
from qiskit.circuit.library import ZFeatureMap
from qiskit.utils import QuantumInstance, algorithm_globals

from qiskit_machine_learning.kernels import QuantumKernel

algorithm_globals.random_seed = 12345

pegasos_backend = QuantumInstance(
    BasicAer.get_backend("statevector_simulator"),
    seed_simulator=algorithm_globals.random_seed,
    seed_transpiler=algorithm_globals.random_seed,
)

feature_map = ZFeatureMap(feature_dimension=num_qubits, reps=1)
qkernel = QuantumKernel(feature_map=feature_map, quantum_instance=pegasos_backend)

from qiskit_machine_learning.algorithms import PegasosQSVC

pegasos_qsvc = PegasosQSVC(quantum_kernel=qkernel, C=C, num_steps=tau)
pegasos_qsvc.fit(train_features, train_labels)
```

yields the error

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
<ipython-input-23-5efd7aedaf88> in <module>
      4
      5 # training
----> 6 pegasos_qsvc.fit(train_features, train_labels)
      7
      8 # testing
```

Assignees
No one assigned

Labels
type: bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Quantum kernel fix for dimensions other than 2
qiskit-community/qiskit-machine-learning

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Participants

```
~/.local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in fit(sel
    184             value = self._compute_weighted_kernel_sum(i, X, training=True)
    185
--> 186             if (self._label_map[y[i]] * self.C / step) * value < 1:
    187                 # only way for a component of alpha to become non zero
    188                 self._alphas[i] = self._alphas.get(i, 0) + 1

ValueError: The truth value of an array with more than one element is ambiguous. Use a.any() or a.all()
```

while substituting `n_features=4` with `n_features=7` and changing the corresponding `num_qubits` to 7 yields the error.

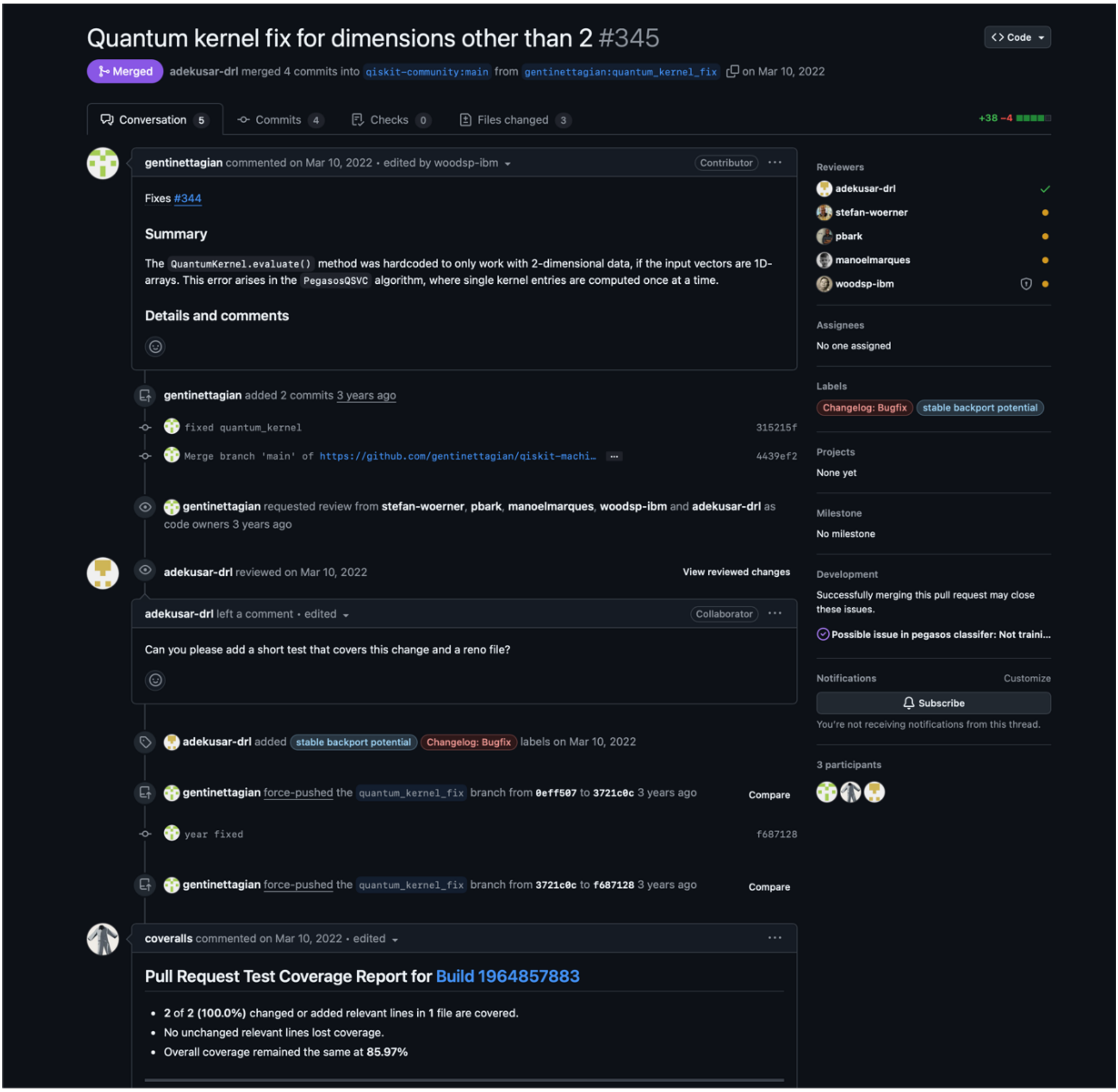
```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
<ipython-input-29-5efd7aedaf88> in <module>
      4
      5 # training
----> 6 pegasos_qsvc.fit(train_features, train_labels)
      7
      8 # testing

~/.local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in fit(sel
    182             i = algorithm_globals.random.integers(0, len(y))
    183
--> 184             value = self._compute_weighted_kernel_sum(i, X, training=True)
    185
    186             if (self._label_map[y[i]] * self.C / step) * value < 1:

~/.local/lib/python3.7/site-packages/qiskit_machine_learning/algorithms/classifiers/pegasos_qsvc.py in _comput
    266                 # evaluate kernel function only for the fixed datum and the data with non zero alpha
    267                 kernel[(index, j)] = kernel.get(
--> 268                     (index, j), self._quantum_kernel.evaluate(X[index], x_j)
    269                 )
    270                 kernel_value = kernel[(index, j)]

~/.local/lib/python3.7/site-packages/qiskit_machine_learning/kernels/quantum_kernel.py in evaluate(self, x_vec
    408
    409         if x_vec.ndim == 1:
--> 410             x_vec = np.reshape(x_vec, (-1, 2))
    411
    412         if y_vec is not None and y_vec.ndim > 2:

<__array_function__ internals> in reshape(*args, **kwargs)

~/.local/lib/python3.7/site-packages/numpy/core/fromnumeric.py in reshape(a, newshape, order)
    296            [5, 6]])
    297     """
--> 298     return _wrapfunc(a, 'reshape', newshape, order=order)
    299
    300

~/.local/lib/python3.7/site-packages/numpy/core/fromnumeric.py in _wrapfunc(obj, method, *args, **kwds)
     55
     56     try:
---> 57         return bound(*args, **kwds)
     58     except TypeError:
     59         # A TypeError occurs if the object does have such a method in its

ValueError: cannot reshape array of size 7 into shape (2)
```

What should happen?

Expected output would be either a working pegasusvqc.fit method or an updated documentation where the num_dimensions is shown as 2.

Any suggestions?

No response



Quantum kernel fix for dimensions other than 2 #345

Code

Merged adekusar-drl merged 4 commits into `qiskit-community:main` from `gentinettagian:quantum_kernel_fix` on Mar 10, 2022

Conversation 5 | Commits 4 | Checks 0 | Files changed 3 | +38 −4

gentinettagian commented on Mar 10, 2022 • edited by woodsp-ibm — Contributor

Fixes #344

Summary

The `QuantumKernel.evaluate()` method was hardcoded to only work with 2-dimensional data, if the input vectors are 1D-arrays. This error arises in the `PegasosQSVC` algorithm, where single kernel entries are computed once at a time.

Details and comments

gentinettagian added 2 commits 3 years ago

`fixed quantum_kernel` 315215f

`Merge branch 'main' of https://github.com/gentinettagian/qiskit-machi…` 4439ef2

gentinettagian requested review from **stefan-woerner**, **pbark**, **manoelmarques**, **woodsp-ibm** and **adekusar-drl** as code owners 3 years ago

adekusar-drl reviewed on Mar 10, 2022 — View reviewed changes

adekusar-drl left a comment • edited — Collaborator

Can you please add a short test that covers this change and a reno file?

adekusar-drl added stable backport potential, Changelog: Bugfix labels on Mar 10, 2022

gentinettagian force-pushed the `quantum_kernel_fix` branch from **0eff507** to **3721c0c** 3 years ago — Compare

`year fixed` f687128

gentinettagian force-pushed the `quantum_kernel_fix` branch from **3721c0c** to **f687128** 3 years ago — Compare

coveralls commented on Mar 10, 2022 • edited

Pull Request Test Coverage Report for Build 1964857883

- **2** of **2 (100.0%)** changed or added relevant lines in **1** file are covered.
- No unchanged relevant lines lost coverage.
- Overall coverage remained the same at **85.97%**

Reviewers
- adekusar-drl
- stefan-woerner
- pbark
- manoelmarques
- woodsp-ibm

Assignees

No one assigned

Labels

Changelog: Bugfix | stable backport potential

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

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3 participants